Linear Algebra

Week 4

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1 Linear systems and elementary operations

A linear system describes several linear conditions that must hold at the same time. Our aim is both computational and conceptual: we want a reliable method for finding every solution, and we want to understand why a system can have no solution, one solution, or a family of solutions. Matrices provide a compact language for this work.

1.1 Writing a system as a matrix equation

Example 1.1 (Three unknowns, one matrix equation)**.** Consider the system over $\mathbb{R}$

$$\begin{cases} x + y + z = 2, \\ x - y + z = 0, \\ y - 2z = 3. \end{cases}$$

It can be written as

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}}_{\mathbf{b}}.$$

Indeed, the three entries of the product are

$$1x + 1y + 1z, \qquad 1x + (-1)y + 1z, \qquad 0x + 1y + (-2)z.$$

Thus the matrix equation contains exactly the original three equations. Direct substitution shows that $(x, y, z) = (2, 1, -1)$ is a solution.

Definition 1.1 (A linear system)**.** A system of m linear equations in n unknowns over a field $\mathbb{F}$ is a system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \qquad\qquad\qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m, \end{cases}$$

where all a_{ij} and b_i are prescribed elements of $\mathbb{F}$, and the unknowns x_j also take values in $\mathbb{F}$. A solution is a choice of all the unknowns that satisfies every equation.

Put $A = (a_{ij}) \in \mathbb{F}^{m \times n}$, $\mathbf{x} = (x_1, \ldots, x_n)^t$, and $\mathbf{b} = (b_1, \ldots, b_m)^t$. Then the system is

$$A_{m \times n}\mathbf{x}_{n \times 1} = \mathbf{b}_{m \times 1}.$$

The matrix A is the *coefficient matrix*; $\mathbf{x}$ is the *vector of unknowns*; and $\mathbf{b}$ is the *right-hand side*, also called the nonhomogeneous part. The vector of unknowns becomes a solution vector only when its entries satisfy the equation.

Definition 1.2 (Homogeneous and nonhomogeneous systems)**.** The system $A\mathbf{x} = \mathbf{b}$ is *homogeneous* when $\mathbf{b} = \mathbf{0}$, and *nonhomogeneous* otherwise. The homogeneous system always has the zero solution.

Definition 1.3 (The augmented matrix)**.** The augmented matrix of $A\mathbf{x} = \mathbf{b}$ is

$$(A \mid \mathbf{b}) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right).$$

The vertical line reminds us that the final column has a different role from the coefficient columns. It does not change the fact that this is an $m \times (n + 1)$ matrix.

Example 1.2 (A flow balance)**.** Suppose three pipes carry flows x, y, z. Measurements give a total inflow of 12, a difference $x - y = 2$, and a balance $y + z = 5$. We obtain

$$\begin{cases} x + y + z = 12, \\ x - y = 2, \\ y + z = 5, \end{cases} \qquad (A \mid \mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 12 \\ 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & 5 \end{array}\right).$$

Subtracting the third equation from the first gives $x = 7$. Then $y = 5$ and $z = 0$. The matrix method developed below automates exactly this kind of elimination.

1.2 The three elementary row operations

Operations on equations should preserve all solutions. Multiplying an equation by a nonzero scalar, exchanging two equations, and adding a multiple of one equation to another all have this property.

Definition 1.4 (Elementary row operations)**.** Let $A = (a_{kl}) \in \mathbb{F}^{m \times n}$, with rows $R_1, \ldots, R_m$. The following are elementary row operations.

1. **Scaling:** $R_i \leftarrow \alpha R_i$, where $\alpha \in \mathbb{F}$ and $\alpha \neq 0$. The resulting matrix $\widetilde{A}$ satisfies

$$\widetilde{a}_{kl} = \begin{cases} a_{kl}, & k \neq i, \\ \alpha a_{il}, & k = i. \end{cases}$$

2. **Interchange:** $R_i \leftrightarrow R_j$, where $i \neq j$. Here

$$\widetilde{a}_{kl} = \begin{cases} a_{kl}, & k \notin \{i, j\}, \\ a_{jl}, & k = i, \\ a_{il}, & k = j. \end{cases}$$

3. **Row replacement:** $R_i \leftarrow R_i + \alpha R_j$, where $i \neq j$ and $\alpha \in \mathbb{F}$. Here

$$\widetilde{a}_{kl} = \begin{cases} a_{kl}, & k \neq i, \\ a_{il} + \alpha a_{jl}, & k = i. \end{cases}$$

These operations make sense for every matrix, whether or not we are currently interpreting it as an augmented matrix. When working on a system, each operation must be applied to the entire augmented row, including the right-hand side.

Example 1.3 (Simplifying the first system)**.** For our opening example,

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & 3 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & 0 & -2 \\ 0 & 1 & -2 & 3 \end{array}\right).$$

Now scale row 2 and eliminate its entry from row 3:

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2} R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 3 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 2 \end{array}\right).$$

The last two rows give $y = 1$ and $z = -1$; the first then gives $x = 2$.

Theorem 1.5 (Every elementary row operation is reversible)**.** The inverse operations are

operation	inverse operation
$R_i \leftarrow \alpha R_i \quad (\alpha \neq 0)$	$R_i \leftarrow \alpha^{-1} R_i$
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$
$R_i \leftarrow R_i + \alpha R_j$	$R_i \leftarrow R_i - \alpha R_j$

In particular, each inverse is again elementary.

Proof. Scaling followed by scaling back multiplies a row by $\alpha^{-1}\alpha = 1$. Interchanging twice restores the order. In row replacement the source row R_j does not change, so $(R_i + \alpha R_j) - \alpha R_j = R_i$. □

Definition 1.6 (Row equivalence)**.** Two matrices A, B of the same size are *row equivalent* if B can be obtained from A by finitely many elementary row operations. We write $A \sim B$.

Theorem 1.7. Row equivalence is an equivalence relation: $A \sim A$; if $A \sim B$, then $B \sim A$; and if $A \sim B$ and $B \sim C$, then $A \sim C$.

Proof. Use no operations for reflexivity, reverse the operations for symmetry, and concatenate the two finite sequences for transitivity. □

Theorem 1.8 (Row operations preserve a system's solution set)**.** Row-equivalent augmented matrices, with the same division into coefficient columns and right-hand side, represent systems with exactly the same solutions.

Proof. Each elementary operation produces an equation satisfied by every old solution. The inverse operation is also elementary, so every solution of the new system is a solution of the old one as well. Repeat this argument along the finite sequence of operations. □

Remark 1.9 (Column operations)**.** One can define elementary column operations and column equivalence in the same way, writing C_i in place of R_i. However, column operations generally change the solution set when the names and order of the unknowns remain fixed. For example, swapping the coefficient columns in $x + 2y = 3$ gives $2x + y = 3$. The pair $(3, 0)$ solves the first equation but not the second. Column operations need a separate interpretation in terms of changes of variables.

2 Echelon forms and Gaussian elimination

2.1 Row echelon form and rank

Definition 2.1 (Leading entry and row echelon form)**.** The *leading entry* of a nonzero row is its first nonzero entry, read from the left. A zero row has no leading entry. A matrix is in *row echelon form*, abbreviated REF, if:

1. all zero rows occur below all nonzero rows;
2. the leading entry of each successive nonzero row lies strictly farther to the right.

In particular, entries below a leading entry are zero. Leading entries need not equal 1 in REF.

Example 2.1. The matrix

$$\begin{pmatrix} 2 & -1 & 4 & 0 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is in REF. The matrix

$$\begin{pmatrix} 1 & 2 & 0 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

is not: the first two leading entries are in the same column. Also, a zero row placed above a nonzero row violates REF.

Row reduction means applying finitely many elementary row operations to reach an echelon matrix. There may be many possible echelon matrices for a given starting matrix. The next result ensures that they agree on an important numerical quantity.

Theorem 2.2 (The number of zero rows in echelon form)**.** If row-equivalent matrices U, V are both in REF, they have the same number of zero rows, and hence the same number of nonzero rows.

The requirement that both matrices be in REF matters: row-equivalent matrices that are not both in REF may have different numbers of zero rows.

Definition 2.3 (Rank)**.** For $A \in \mathbb{F}^{m\times n}$, its *rank*, denoted $\operatorname{rank}(A)$ or $r(A)$, is the number of nonzero rows in any REF matrix row equivalent to A.

The preceding theorem makes this definition independent of the chosen row reduction.

Corollary 2.4. Row-equivalent matrices have equal rank. Moreover,

$$\operatorname{rank}(0_{m\times n}) = 0, \qquad \operatorname{rank}(I_n) = n.$$

Theorem 2.5 (Bounds on rank)**.** For $A \in \mathbb{F}^{m\times n}$,

$$0 \le \operatorname{rank}(A) \le \min\{m, n\}.$$

A matrix has *full rank* if its rank is $\min\{m, n\}$.

Proof. There are at most m nonzero rows. Their leading entries occupy distinct columns, so there are also at most n of them. □

2.2 Reduced row echelon form

Definition 2.6 (Reduced row echelon form)**.** A matrix is in *reduced row echelon form*, abbreviated RREF, if:

1. it is in REF;
2. every leading entry equals 1;
3. each leading entry is the only nonzero entry in its column.

The leading entries are often called *pivots*, and their columns are called *pivot columns*.

Example 2.2. The matrix

$$\begin{pmatrix} 1 & 0 & -2 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

is in RREF. The third and fifth columns need not have the form of identity columns because they contain no pivots. By contrast,

$$\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is in REF but not RREF: the last pivot has nonzero entries above it.

Theorem 2.7 (Existence and uniqueness of RREF)**.** Every matrix over a field is row equivalent to exactly one matrix in RREF.

The existence part is constructive: the following algorithm finds the matrix.

Gaussian elimination, followed by reduction above the pivots.

1. Start with the entire matrix as the active part. If it is zero, stop.
2. Find the leftmost column containing a nonzero entry in the active part. Move a row containing such an entry to the top of the active part.
3. Multiply that row by the inverse of its leading entry, so the pivot becomes 1.
4. Subtract suitable multiples of this pivot row from every row below it to make the entries below the pivot zero.
5. Apply the same procedure to the rows below the pivot, ignoring columns up to and including its pivot column. Each new pivot lies strictly to the right of the previous pivot. Continue until only zero rows remain. This gives REF.
6. Work upward from the last pivot. Use each pivot row to eliminate the entries above its pivot. The result is RREF.

Every division in this algorithm is by a nonzero field element. This is why the same method works over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and finite fields such as $\mathbb{Z}_5$.

2.3 A complete elimination example

Example 2.3 (Five equations in four unknowns)**.** Consider the system over $\mathbb{R}$ whose augmented matrix is

$$M_0 = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 2 & 3 & 4 & 1 & 10 \\ 1 & 0 & 0 & 4 & 9 \\ 3 & 5 & 6 & 0 & 11 \\ 7 & 9 & 11 & 6 & 34 \end{array}\right).$$

For example, its second equation is $2x + 3y + 4z + w = 10$. First use row 1 to eliminate the entries below its pivot:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1,$$
$$R_4 \leftarrow R_4 - 3R_1, \quad R_5 \leftarrow R_5 - 7R_1.$$

These operations give

$$M_1 = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & -1 & 2 \\ 0 & -1 & -1 & 3 & 5 \\ 0 & 2 & 3 & -3 & -1 \\ 0 & 2 & 4 & -1 & 6 \end{array}\right).$$

Next perform $R_3 \leftarrow R_3 + R_2$, $R_4 \leftarrow R_4 - 2R_2$, and $R_5 \leftarrow R_5 - 2R_2$:

$$M_2 = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & -1 & 2 \\ 0 & 0 & 1 & 2 & 7 \\ 0 & 0 & -1 & -1 & -5 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right).$$

Now $R_4 \leftarrow R_4 + R_3$, followed by $R_5 \leftarrow R_5 - R_4$, produces REF:

$$U = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & -1 & 2 \\ 0 & 0 & 1 & 2 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Back-substitution already solves the system:

$$\begin{aligned} w &= 2, \\ z + 2w = 7 &\implies z = 3, \\ y + 2z - w = 2 &\implies y = 2 - 6 + 2 = -2, \\ x + y + z + w = 4 &\implies x = 4 + 2 - 3 - 2 = 1. \end{aligned}$$

To obtain RREF, first use row 4 to clear column 4 above its pivot:

$$R_3 \leftarrow R_3 - 2R_4, \quad R_2 \leftarrow R_2 + R_4, \quad R_1 \leftarrow R_1 - R_4.$$

The nonzero rows become

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right).$$

Finally apply $R_2 \leftarrow R_2 - 2R_3$, $R_1 \leftarrow R_1 - R_3$, and $R_1 \leftarrow R_1 - R_2$:

$$\mathrm{RREF}(M_0) = \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

The unique solution is $(1, -2, 3, 2)$. The last zero row records a redundant equation; it does not create a free variable because every unknown has a pivot.

3 Reading the solution set

3.1 Contradictions and free variables

Example 3.1 (Detecting inconsistency). Over $\mathbb{R}$, consider

$$\begin{cases} x + 2y + z = 2, \\ 2x - y + z = 1, \\ 3x + y + 2z = 5. \end{cases}$$

The left-hand side of the third equation is the sum of the first two left-hand sides, but its right-hand side is not $2 + 1$. Row reduction makes the contradiction explicit:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & -1 & 1 & 1 \\ 3 & 1 & 2 & 5 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -5 & -1 & -3 \\ 0 & -5 & -1 & -1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -5 & -1 & -3 \\ 0 & 0 & 0 & 2 \end{array}\right).$$

The last row means $0 = 2$, so there is no solution. A row of the form $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ is called a *contradiction row*.

Example 3.2 (A family of solutions)**.** Over $\mathbb{R}$, consider

$$\begin{cases} x + y + z = 4, \\ x - y - z = 2. \end{cases}$$

Its augmented matrix reduces as follows:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & -1 & -1 & 2 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & -2 & -2 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 1 & 1 \end{array}\right).$$

Thus $x = 3$ and $y + z = 1$. Setting $z = t$, every solution has the form

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 - t \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Here x, y are pivot variables and z is a free variable.

Definition 3.1 (Pivot variables, free variables, and degrees of freedom)**.** After reducing a consistent augmented matrix to RREF, an unknown whose coefficient column contains a pivot is a *pivot variable*, also called a bound variable. The remaining unknowns are *free variables*. The number of free variables is the number of *degrees of freedom* of the system.

Each free variable can be chosen independently in $\mathbb{F}$. The equations then determine all pivot variables uniquely.

3.2 The rank criterion

When we row reduce $(A \mid \mathbf{b})$, the operations simultaneously row reduce the coefficient part A. There may be a row that is zero in the coefficient part but nonzero in the final column. Consequently,

$$\operatorname{rank}(A) \leq \operatorname{rank}(A \mid \mathbf{b}).$$

Theorem 3.2 (Consistency criterion)**.** The system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if

$$\operatorname{rank}(A) = \operatorname{rank}(A \mid \mathbf{b}).$$

Proof. If a contradiction row occurs, the corresponding system has no solution, and the augmented matrix has one more nonzero row in echelon form than the coefficient part. If there is no contradiction row, every nonzero echelon row has its pivot in the coefficient part. The two ranks are equal, and back-substitution, after freely choosing the nonpivot variables, produces a solution. Thus equality is equivalent to consistency. □

Corollary 3.3 (Counting the free variables)**.** A consistent system in n unknowns has

$$n - \operatorname{rank}(A)$$

free variables. In particular, it has a unique solution if and only if

$$\operatorname{rank}(A) = \operatorname{rank}(A \mid \mathbf{b}) = n.$$

Proof. There is one pivot variable for each nonzero echelon row of A. All other variables are free. There are no choices left precisely when their number is zero. □

Example 3.3 (A parameter controls consistency). For $a \in \mathbb{R}$, consider

$$\begin{cases} x + 2y = 3, \\ 2x + 4y = a. \end{cases} \qquad \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & a \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & a-6 \end{array}\right).$$

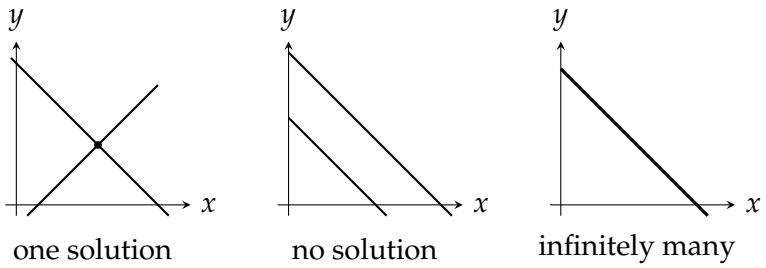
If $a \neq 6$, there is a contradiction. If $a = 6$, there is one free variable, and the solutions are $(x, y) = (3 - 2t, t)$ with $t \in \mathbb{R}$.

3.3 How many solutions can there be?

Theorem 3.4 (Number of solutions over an infinite field). If $\mathbb{F}$ is infinite, a linear system has either no solution, exactly one solution, or infinitely many solutions. A homogeneous system cannot have no solution.

Proof. A contradiction gives no solution. Without a contradiction, zero free variables gives one solution. With at least one free variable, varying it over the infinite field gives infinitely many distinct solutions. The zero vector always solves a homogeneous system. □

Over $\mathbb{R}$, two equations in two unknowns can describe two lines that meet once, two distinct parallel lines, or the same line. The corresponding solution sets contain one point, no points, or an entire line.



one solution no solution infinitely many

Remark 3.5 (Finite fields). The infinite-field assumption is essential. Over a finite field with q elements, a consistent system with d free variables has exactly q^d solutions: there are q independent choices for each free variable.

Example 3.4 (Solving over $\mathbb{Z}_5$). Solve

$$\begin{cases} x + 2y = 4, \\ 2x + y = 0 \end{cases} \qquad \text{over } \mathbb{Z}_5.$$

All entries and row operations are interpreted modulo 5:

$$\left(\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 1 & 0 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 2 & 2 \end{array}\right) \xrightarrow{R_2 \leftarrow 3R_2} \left(\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right).$$

We used $2^{-1} = 3$ in $\mathbb{Z}_5$, since $2 \times 3 \equiv 1 \pmod 5$. The unique solution is $(2, 1)$, and $2 \times 2 + 1 = 5 = 0$ in this field.

For comparison, $x + y = 1$ over $\mathbb{Z}_5$ has one free variable and exactly five solutions:

$$(1, 0),\ (0, 1),\ (4, 2),\ (3, 3),\ (2, 4).$$

3.4 A particular solution plus a homogeneous solution

Theorem 3.6 (Structure of a nonhomogeneous solution set)**.** Let $A \in \mathbb{F}^{m\times n}$, $\mathbf{b} \in \mathbb{F}^m$, and write

$$W = \{\mathbf{h} \in \mathbb{F}^n : A\mathbf{h} = \mathbf{0}\}.$$

If $\mathbf{x}_p$ is one solution of $A\mathbf{x} = \mathbf{b}$, then its complete solution set is

$$\{\mathbf{x}_p + \mathbf{h} : \mathbf{h} \in W\}.$$

Proof. For $\mathbf{h} \in W$, distributivity gives

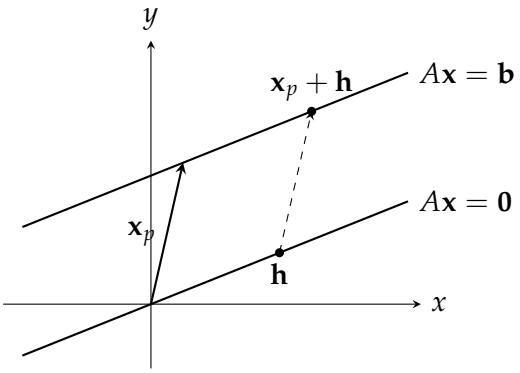
$$A(\mathbf{x}_p + \mathbf{h}) = A\mathbf{x}_p + A\mathbf{h} = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$

Conversely, if $A\mathbf{x} = \mathbf{b}$, set $\mathbf{h} = \mathbf{x} - \mathbf{x}_p$. Then

$$A\mathbf{h} = A\mathbf{x} - A\mathbf{x}_p = \mathbf{b} - \mathbf{b} = \mathbf{0},$$

so $\mathbf{h} \in W$ and $\mathbf{x} = \mathbf{x}_p + \mathbf{h}$. □

In words, first find one point satisfying the measurements. Then add every change that leaves the homogeneous measurements equal to zero. Geometrically, this translates the homogeneous solution set by $\mathbf{x}_p$.



The picture illustrates one degree of freedom over $\mathbb{R}$; the theorem itself holds in $\mathbb{F}^n$ for every field.

Example 3.5. For $x + y + z = 4$, $x - y - z = 2$, we found

$$\mathbf{x}_p = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{h} = t \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Indeed, the homogeneous equations $x + y + z = 0$ and $x - y - z = 0$ give $x = 0$ and $y = -z$. Thus the entire homogeneous solution set is exactly the family of vectors $\mathbf{h}$ above, and adding $\mathbf{x}_p$ gives every nonhomogeneous solution.

4 Invertible matrices

4.1 Definition and first criteria

Example 4.1 (An inverse need not exist)**.** For a diagonal matrix over $\mathbb{R}$,

$$\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = I_2.$$

However, if $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, no matrix B can satisfy $AB = I_2$. Every product AB has its second row equal to twice its first row, whereas I_2 does not.

Definition 4.1 (An invertible matrix). A square matrix $A \in \mathbb{F}^{n\times n}$ is *invertible* if there exists $B \in \mathbb{F}^{n\times n}$ such that $AB = I_n$. A square matrix that is not invertible is called *singular*.

This definition initially asks only for a right inverse. For square matrices over a field, a right inverse is also a left inverse.

Theorem 4.2 (Four equivalent conditions). For $A \in \mathbb{F}^{n\times n}$, the following are equivalent:

1. A is row equivalent to I_n;
2. $\operatorname{rank}(A) = n$;
3. for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has exactly one solution;
4. for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has at least one solution.

Proof. If $A \sim I_n$, its rank is n. Conversely, an $n \times n$ RREF matrix of rank n has a pivot in every row and column, so it is I_n. This proves the equivalence of (1) and (2).

If $\operatorname{rank}(A) = n$, then the augmented matrix has rank at least n and at most its number of rows, which is n. Hence the ranks are equal and all n variables are pivot variables. The system has exactly one solution for every right-hand side. Thus (2) implies (3), and (3) immediately implies (4).

To prove that (4) implies (2), row reduce A by a fixed sequence of operations. If its rank were less than n, its echelon form would have a zero row. Choose a transformed right-hand side with entry 1 in that row. Reversing the same operations on this right-hand side produces some original $\mathbf{b}$. For that $\mathbf{b}$ the transformed augmented matrix has a contradiction row, violating (4). Therefore the rank is n. □

Theorem 4.3 (A left and a right inverse agree). If $AB = I_n$ and $CA = I_n$, then $B = C$.

Proof. Associativity gives $C = C(AB) = (CA)B = B$. □

Theorem 4.4 (A right inverse of a square matrix is also a left inverse). If $A, B \in \mathbb{F}^{n\times n}$ and $AB = I_n$, then $\operatorname{rank}(A) = n$ and $BA = I_n$.

First step. For any $\mathbf{b} \in \mathbb{F}^n$, the choice $\mathbf{x} = B\mathbf{b}$ satisfies

$$A\mathbf{x} = A(B\mathbf{b}) = (AB)\mathbf{b} = I_n\mathbf{b} = \mathbf{b}.$$

The fourth condition in Theorem 4.2 holds, so $\operatorname{rank}(A) = n$ and $A \sim I_n$. We complete the proof after describing row operations by matrix multiplication. □

4.2 Elementary matrices

Example 4.2 (A row operation as multiplication). For

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix},$$

the operation $R_2 \leftarrow R_2 - \frac{3}{2}R_1$ gives

$$\begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Apply the same operation to I_2:

$$I_2 \longmapsto E = \begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}.$$

Multiplication reproduces the row operation:

$$EA = \begin{pmatrix} 1\times 2+0\times 3 & 1\times 1+0\times 2 \\ -\frac{3}{2}\times 2+1\times 3 & -\frac{3}{2}\times 1+1\times 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Definition 4.5 (Elementary matrix)**.** The elementary matrix E_θ corresponding to a row operation θ on m rows is obtained by applying θ to I_m.

Theorem 4.6 (Left multiplication performs a row operation)**.** If $A \in \mathbb{F}^{m\times n}$, then $E_\theta A$ is the matrix obtained by applying θ to A.

Proof. For row scaling, E_θ equals I_m except that its (i,i) entry is α. Thus

$$(E_\theta A)_{kl} = \sum_{p=1}^{m} (E_\theta)_{kp} a_{pl} = \begin{cases} a_{kl}, & k \neq i, \\ \alpha a_{il}, & k = i. \end{cases}$$

For a swap, row i of E_θ has its single 1 in column j, and row j has its single 1 in column i; the same sum therefore selects a_{jl} and a_{il} respectively. Other rows remain unchanged.

For $R_i \leftarrow R_i + \alpha R_j$, row i of E_θ has entries 1 in column i and α in column j, and all its other entries are zero. Hence the ith row of the product has entries

$$\sum_{p=1}^{m} (E_\theta)_{ip} a_{pl} = a_{il} + \alpha a_{jl}.$$

All other rows are unchanged. These are exactly the three elementary operations. □

Remark 4.7. Elementary column operations can be represented by multiplication on the right, using the matrix obtained by performing the corresponding column operation on an identity matrix.

Corollary 4.8. Two $m \times n$ matrices A, B are row equivalent if and only if

$$B = E_k \cdots E_2 E_1 A$$

for a finite sequence of elementary $m \times m$ matrices. The rightmost factor acts first.

Completion of the proof of Theorem 4.4. Since $A \sim I_n$, there are elementary matrices such that

$$CA = I_n, \qquad C = E_k \cdots E_1.$$

Together with $AB = I_n$, the left-right inverse result gives $C = B$. Therefore $BA = CA = I_n$. □

Corollary 4.9 (Uniqueness of the inverse)**.** An invertible matrix A has a unique inverse. It is denoted by A^{-1} and satisfies

$$AA^{-1} = A^{-1}A = I_n.$$

For every $\mathbf{b}$, the unique solution of $A\mathbf{x} = \mathbf{b}$ is

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

Proof. If $AB = I_n$ and $AD = I_n$, the theorem gives $BA = I_n$, and $B = B(AD) = (BA)D = D$. Multiplying $A\mathbf{x} = \mathbf{b}$ on the left by A^{-1} gives the stated solution formula. □

4.3 Finding an inverse by row reduction

Suppose a sequence of row operations sends A to I_n. Its elementary matrices satisfy

$$E_k \cdots E_1 A = I_n.$$

Consequently $E_k \cdots E_1 = A^{-1}$. Applying the same operations, in the same order, to I_n therefore gives A^{-1}. We can perform both calculations simultaneously:

$$(A \mid I_n) \longrightarrow (I_n \mid A^{-1}).$$

If row reduction shows that the left block has rank less than n, the matrix has no inverse.

Example 4.3 (A 2×2 inverse). Let $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$. Over $\mathbb{R}$,

$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 3 & 2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{3}{2} & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow 2R_2} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -3 & 2 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - \frac{1}{2}R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array}\right).$$

Thus $A^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. An entry-by-entry check gives

$$AA^{-1} = \begin{pmatrix} 2\times 2 + 1 \times (-3) & 2 \times (-1) + 1 \times 2 \\ 3 \times 2 + 2 \times (-3) & 3 \times (-1) + 2 \times 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$A^{-1}A = \begin{pmatrix} 2\times 2 + (-1) \times 3 & 2 \times 1 + (-1) \times 2 \\ (-3) \times 2 + 2 \times 3 & (-3) \times 1 + 2 \times 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Example 4.4 (A 3×3 inverse). Let

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The augmented calculation is

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 3 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 6 & -3 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right).$$

Then clear the entries above the third pivot:

$$\begin{array}{c} R_1 \leftarrow R_1 + 2R_3 \\ R_2 \leftarrow R_2 - 6R_3 \end{array} \xrightarrow{\quad} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & -3 & 1 & -6 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right).$$

Therefore

$$A^{-1} = \begin{pmatrix} 1 & 0 & 2 \\ -3 & 1 & -6 \\ 0 & 0 & 1 \end{pmatrix}.$$

For example, entry $(2,3)$ of AA^{-1} is $3 \times 2 + 1 \times (-6) + 0 \times 1 = 0$, as required.

4.4 Properties of inverses

Theorem 4.10. For invertible matrices of the indicated sizes:

1. A^{-1} is invertible and $(A^{-1})^{-1} = A$;
2. A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$;
3. if $A, B \in \mathbb{F}^{n\times n}$ are invertible, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Proof. The first statement follows from $A^{-1}A = AA^{-1} = I$. For the second, transpose both inverse identities and use the reversal of order under transposition:

$$(A^{-1})^t A^t = (AA^{-1})^t = I, \qquad A^t(A^{-1})^t = (A^{-1}A)^t = I.$$

For the third, compute both orders:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$$

Thus the inverse of a product reverses the order of its factors. □

Theorem 4.11 (Inverses of elementary matrices)**.** Every elementary matrix is invertible. Its inverse is the elementary matrix for the inverse row operation.

Proof. Performing a row operation and its inverse restores every matrix, in particular I. By the multiplication interpretation, the two elementary matrices multiply to I in both orders. □

For instance, the inverse of $\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$ is $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$: the operations add $-3R_1$ and $3R_1$ to row 2, respectively.

4.5 The complete equivalence theorem

Theorem 4.12 (Equivalent descriptions of invertibility)**.** For a square matrix $A \in \mathbb{F}^{n\times n}$, the following are equivalent:

1. A is invertible;
2. A is a product of finitely many elementary matrices;
3. A is row equivalent to I_n;
4. $\operatorname{rank}(A) = n$;
5. for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has exactly one solution;
6. for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a solution.

Proof. Conditions (3)–(6) were proved equivalent in Theorem 4.2. If A is invertible, $A\mathbf{x} = \mathbf{b}$ is solved by $A^{-1}\mathbf{b}$, so (1) implies (6).

If (3) holds, there are elementary matrices with $E_k \cdots E_1 A = I$. Each is invertible, so

$$A = E_1^{-1}E_2^{-1}\cdots E_k^{-1}.$$

Every inverse on the right is elementary, giving (2). Finally, a product of invertible elementary matrices is invertible, so (2) implies (1). □

Corollary 4.13 (Row equivalence and invertible left multiplication)**.** Let $A, B \in \mathbb{F}^{m\times n}$. Then $A \sim B$ if and only if there is an invertible $P \in \mathbb{F}^{m\times m}$ such that

$$B = PA.$$

In particular, whenever P is invertible,

$$\operatorname{rank}(PA) = \operatorname{rank}(A).$$

Proof. If $A \sim B$, write $B = E_k \cdots E_1 A$ and take $P = E_k \cdots E_1$, which is invertible. Conversely, an invertible P is a product of elementary matrices, so PA is obtained from A by row operations. The rank equality follows because row-equivalent matrices have equal rank. □